Simple Linear Regression

Lab (7)

1. House Price Prediction

- **Context: Predict house prices based on house size (simulated data).**
  X: house_size (sq. ft.)
  Y: price ($)
- **Tasks:**
  - Generate synthetic data: 100 samples with house_size (1,000–5,000 sq. ft.) and price ($100k–$500k).
  - Fit a linear regression model.
  - Predict the price for a 3,200 sq. ft. house.
- **Questions:**
  - What is the slope? Interpret it in context.
  - What price does the model predict for a 0 sq. ft. lot? Is this realistic?

2. Exam Scores vs. Study Hours

- **Context: Explore the relationship between study time and exam scores.**
  X: study_hours (hours/day)
  Y: exam_score (0–100)

```
# Data Generation
   np.random.seed(42)
   study_hours = np.random.uniform(1, 10, 50).reshape(-1, 1)
   exam_score = 50 + 5*study_hours + np.random.normal(0, 8, 50).reshape(-1, 1)
```

- **Tasks:**
  - Create data: 50 samples with study_hours (1–10) and exam_score (50–100).
  - Split data (80% train, 20% test).
  - Train the model and evaluate using MSE.

- **Questions:**

  - How much does the score increase per additional study hour?
  - What score is predicted for 5 hours of study?

3. Diabetes Progression Prediction

- **Context: Predict disease progression using BMI (built-in dataset).**
  X: BMI (bmi, scaled)
  Y: Disease progression (target)

```
# Load Dataset
   diabetes = load_diabetes()
```

- **Tasks:**

  - Load the diabetes dataset: load_diabetes().
  - Use only the bmi feature.
  - Plot data and regression line.

- **Questions:**

  - What is the R^2 score? Is the model useful?
  - How does progression change for a BMI increase of 1 unit?

4. Temperature vs. Ice Cream Sales

- **Context: Model ice cream sales based on temperature.**
  X: temperature (°C)
  Y: sales (units sold)

```
# Data Generation
    np.random.seed(42)
    temperature = np.random.uniform(15, 35, 50).reshape(-1, 1)
    sales = 20 + 5*temperature + np.random.normal(0, 10, 50).reshape(-1, 1)
```

- **Tasks:**

  - Generate data: Temperature (15°C–35°C), sales (50–200 units).
  - Fit the model and calculate MSE.
  - Predict sales at 25°C.

- **Questions:**

- Is the intercept meaningful here? Why?
- What sales are predicted at 0°C? Is this reasonable?

5. Employee Salary vs. Experience

- **Context: Predict salary based on years of experience.**
  X: experience (years)
  Y: salary ($1,000s)

```
# Data Generation
    np.random.seed(42)
    experience = np.random.uniform(0, 20, 100).reshape(-1, 1)
    salary = 40 + 8*experience + np.random.normal(0, 10, 100).reshape(-1, 1)
```

- **Tasks:**

- Create data: Experience (0–20 years), salary (40–200k).
- Split data (70% train, 30% test).
- Evaluate using R^2.

- **Questions:**

- What salary is predicted for 15 years of experience?
- What percentage of salary variation is explained by experience?

6. Car Fuel Efficiency

- **Context: Model MPG based on engine displacement.**
  X: displacement (L)
  Y: mpg (miles/gallon)

```
url = "https://raw.githubusercontent.com/mwaskom/seaborn-data/master/mpg.csv"
    data = pd.read_csv(url)
```

- **Tasks:**

- Use the mtcars dataset (load via URL).
- Plot displacement vs. MPG with regression line.

- **Questions:**

  - Interpret the slope. Is higher displacement linked to better/worse MPG?
  - What MPG is predicted for a 4.0L engine?

7. Product Price vs. Demand

- **Context: Analyze how price affects demand.**
  X: price ($)
  Y: demand (units sold)

```
# Data Generation (inverse relationship)
   np.random.seed(42)
   price = np.random.uniform(10, 50, 50).reshape(-1, 1)
   demand = 300 - 6*price + np.random.normal(0, 15, 50).reshape(-1, 1)
```

- **Tasks:**

  - Generate data: Price ($10–$50), demand (200–20 units).
  - Fit the model and calculate residuals.

- **Questions:**

  - Why is the slope negative?
  - What demand is predicted at $35?

8. Student Height vs. Weight

- **Context: Predict weight using height.**
  X: height (cm)
  Y: weight (kg)

```
# Data Generation
    np.random.seed(42)
    height = np.random.uniform(150, 200, 100).reshape(-1, 1)
    weight = 0.8*height - 70 + np.random.normal(0, 5, 100).reshape(-1, 1)
```

- **Tasks:**

  - Create synthetic data: Height (150–200 cm), weight (45–100 kg).

- Train the model and compute MSE.

- **Questions:**

  - What weight is predicted for 175 cm?
  - Why is the intercept not zero?

9. Advertising Spend vs. Sales

- **Context: Model sales based on TV ad spending.**
  X: tv_ad_budget ($1,000s)
  Y: sales (units)

```
# Data Generation
    np.random.seed(42)
    tv_budget = np.random.uniform(1, 100, 100).reshape(-1, 1)
    sales = 50 + 8*tv_budget + np.random.normal(0, 30, 100).reshape(-1, 1)
```

- **Tasks:**

  - Generate data: TV budget (1–100k), sales (100–1,000 units).
  - Split data (75% train, 25% test).

- **Questions:**

  - How much do sales increase per $1k spent on TV ads?
  - What sales are predicted with a $80k budget?

10. Crop Yield vs. Rainfall

- **Context: Predict crop yield based on rainfall.**
  X: rainfall (mm)
  Y: crop_yield (tons/hectare)

```
# Data Generation
    np.random.seed(42)
    rainfall = np.random.uniform(300, 900, 50).reshape(-1, 1)
    crop_yield = 2 + 0.007*rainfall + np.random.normal(0, 0.5, 50).reshape(-1, 1)
```

- **Tasks:**

  - Generate data: Rainfall (300–900 mm), yield (2–8 tons).

- Plot residuals and check patterns.

- **Questions:**

  - Is the relationship linear?
  - What yield is expected with 600 mm of rain?